

Lecture 1: Calibration and Online Learning

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1 Introduction

The primary topic of this lecture is *calibration*, one of the central notions of reliability in modern machine learning systems. Informally, calibration measures whether predicted probabilities accurately reflect empirical frequencies observed in the underlying data distribution.

For example, if a classifier predicts an event with confidence 0.7, then approximately 70% of all such predictions should correspond to positive outcomes. Consequently, calibration is concerned not only with predictive accuracy, but also with the reliability and interpretability of probabilistic predictions.

The lecture further introduces the framework of *prediction with expert advice*, together with the notions of external regret and internal (swap) regret in online learning.

2 Calibration

Let

$$(x, y) \sim \mathcal{D},$$

where x denotes the input and y denotes the associated label.

A predictor

$$f : \mathcal{X} \rightarrow [0, 1]$$

is said to be *calibrated* if

$$\mathbb{E}[y \mid f(x) = \hat{y}] = \hat{y}.$$

That is, conditioned on all examples receiving prediction $\hat{y}$, the empirical average of the labels equals $\hat{y}$ itself.

2.1 Binary Classification Setting

Let:

$$y \in \{0, 1\}.$$

In this setting, the model outputs

$$f(x) \in [0, 1],$$

which is interpreted as the estimated probability that $y=1$.

3 Examples of Calibration

Consider a sequence of independent fair coin flips:

$$y_1, y_2, \dots, y_T.$$

The objective is to predict the probability of observing heads.

3.1 Perfectly Calibrated Predictor

Suppose the predictor always outputs

$$f(x) = \frac{1}{2}.$$

Since the coin is fair,

$$\mathbb{E}[y] = \frac{1}{2},$$

and therefore

$$\mathbb{E}[y \mid f(x) = 1/2] = 1/2.$$

Hence, the predictor is calibrated.

3.2 Non-Calibrated Predictor

Now suppose that whenever

$$f(x) = 0,$$

the outcomes satisfy

$$y = \begin{cases} 0 & \text{with probability } \frac{2}{3}, \\ 1 & \text{with probability } \frac{1}{3}. \end{cases}$$

Then

$$\mathbb{E}[y \mid f(x) = 0] = \frac{1}{3} \neq 0.$$

Thus, the predictor fails to satisfy calibration.

Similarly, if

$$f(x) = 1,$$

and

$$y = \begin{cases} 0 & \text{with probability } \frac{1}{3}, \\ 1 & \text{with probability } \frac{2}{3}, \end{cases}$$

then

$$\mathbb{E}[y \mid f(x) = 1] = \frac{2}{3} \neq 1.$$

Again, the predictor is not calibrated.

4 Motivation for Calibration

Calibration plays a crucial role in several domains.

4.1 Economics and Decision Theory

In economics and decision theory, agents frequently optimize decisions based on probabilistic forecasts. Reliable confidence estimates are therefore essential for maximizing expected utility and minimizing decision-theoretic risk. Poorly calibrated predictions may lead to systematically sub-optimal decisions, even if the underlying classifier achieves high accuracy.

4.2 Machine Learning

Modern machine learning models, particularly deep neural networks, are often highly accurate but poorly calibrated. Such models commonly exhibit:

- overconfidence,
- underconfidence,
- unreliable uncertainty estimates.

Calibration methods attempt to correct these discrepancies while preserving predictive performance.

5 Calibration Error

Given a predictor f , one may seek a calibrated function g minimizing

$$\min_{g \text{ calibrated}} |f(\cdot) - g(\cdot)|.$$

One commonly used metric is the *Expected Calibration Error (ECE)*, which informally measures the discrepancy between predicted confidence and empirical frequency.

The board presents the quantity

$$\mathbb{E} [|\mathbb{E}[y \mid g] - \hat{y}|].$$

Intuitively, ECE captures the average deviation between:

- predicted probabilities,
- observed outcome frequencies.

6 Prediction with Expert Advice

The lecture then transitions to the online learning framework known as *prediction with expert advice*.

6.1 Setting

Suppose there are n experts or actions. At each round

$$t = 1, 2, \dots, T,$$

the algorithm selects a distribution

$$p^{(t)} \in \Delta^n,$$

where Δ^n denotes the probability simplex over the set of experts.

After the decision is made, a loss vector is revealed

$$\ell^{(t)} \in [0, 1]^n$$

The coordinate

$$\ell_i^{(t)}$$

corresponds to the loss incurred by expert i at time t .

The algorithm therefore incurs expected loss

$$\langle p^{(t)}, \ell^{(t)} \rangle.$$

7 External Regret

The objective is to design an algorithm, where the cumulative loss is nearly as small as that of the best fixed expert chosen in hindsight.

7.1 Minimal External Regret

There exist algorithms satisfying

$$\sum_{t=1}^T \langle p^{(t)}, \ell^{(t)} \rangle \leq \sum_{t=1}^T \ell_j^{(t)} + O(\sqrt{T \log n})$$

for all experts

$$j \in [n].$$

Thus, the regret relative to the best fixed expert grows sublinearly in T , implying that the average regret converges to zero as

$$T \rightarrow \infty.$$

8 Internal Regret

Rather than comparing against a single fixed expert, internal regret compares against strategies that systematically replace one action with another.

Formally, consider mappings

$$m : [n] \rightarrow [n].$$

Such mappings specify that whenever the algorithm selects action i , it should instead have selected action $m(i)$. Internal regret measures the quantity

$$\text{Loss of algorithm} - \min_m \text{Loss}(m(\text{algorithm})).$$

Achieving low internal regret guarantees that no systematic action-swapping strategy would have substantially improved performance.

9 Algorithm for Minimizing Internal Regret

We cover the algorithm of [Blum and Mansour \[2007\]](#), which performs a reduction from internal regret minimization to external regret minimization.

For every action

$$i \in [n],$$

instantiate an external regret minimization algorithm

$$A_i.$$

Each algorithm outputs a distribution

$$q_i^{(t)}.$$

Collect these distributions into a matrix

$$Q^{(t)}.$$

The algorithm then selects a distribution

$$p^{(t)}$$

satisfying the fixed-point condition

$$p^{(t)} = p^{(t)} Q^{(t)}.$$

After observing losses

$$\ell^{(t)},$$

each algorithm A_i is updated using the perceived loss vector

$$p_i^{(t)} \ell^{(t)}.$$

10 Analysis Sketch

For each algorithm A_i , the external regret guarantee implies

$$\sum_{t=1}^T \langle q_i^{(t)}, p_i^{(t)} \ell^{(t)} \rangle \leq \sum_{t=1}^T p_i^{(t)} \ell_j^{(t)} + O(\sqrt{T \log n}).$$

Furthermore,

$$(p^{(t)})^T Q^{(t)} \ell^{(t)} = (p^{(t)})^T \ell^{(t)},$$

which equals the true loss incurred by the algorithm.

Combining these inequalities yields

$$\text{Algorithm Loss} \leq \text{Loss of Best Swapped Policy} + O(\sqrt{T \log n}).$$

Consequently, the procedure achieves low swap regret.

References

Avrim Blum and Yishay Mansour. From external to internal regret. *Journal of Machine Learning Research*, 8(47):1307–1324, 2007. URL <https://www.jmlr.org/papers/volume8/blum07a/blum07a.pdf>.